MTech in Robotics at IIT Delhi

Inter-disciplinary program offered jointly by:

Department of Computer Science & Engineering
Department of Electrical Engineering
Department of Mechanical Engineering
Yardi School of Artificial Intelligence
(in association with the Center of Excellence on Biologically Inspired Robots and Drones)

Admission

GATE **(CS/ME/EE/EC/Aerospace/IN/)** (required for institute fellowship)
Based on interview. (Shortlisting for Interview includes a Programming Test)

Course Credit Structure

Category	PC	PE	OC	Total
Credits	42	6	3	51 + bridge

Bridge (BR: At least one of the following two, but waived for people with preparation in both)

COL671 Artificial Intelligence 3 0 2 (Intro AI course)
ELL225/MCL212 (Intro control course)

Specializations

The study of Robotics is by its nature broad and inter-disciplinary. It is hence common for experts to seek specialization in certain aspects. The program facilitates such specialization, if a student chooses it. Four application-centric specializations are offered:

1. Collaborative Robotics
2. Industrial Robotics
3. Rehabilitation and Medical Robotics
4. Autonomous and Intelligent Vehicles

An interested student may request specialization at any point during their program. To earn a specialization, the student will be required to complete 6 credits of electives designated for that specialization, and undertake projects (P1, P2, and P3) in that specific area. (The projects must be certified by the advisor to be relevant to the specialization sought.) On recommendation by the program executive committee, a certificate of specialization will be provided. A student may choose to not opt for any specialization and complete electives and projects in different specialization areas.

Program Core (42 credits)

Foundational Concepts (FC: 6 credits)

FC1. **JRL704 Robotics Lab 1-0-4**
FC2. **ELL701 Mathematical Methods Incontrol 3-0-0**

Machines, Mechanics and Control (MC: 5 credits)

MC1. **MCL799 Mechanics and Control of Robots 3-0-0**
MC2. **JRL747 Degrees of Freedom and Constraints in Devices 2-0-0**

Autonomous Reasoning and Cognition (RC: 4 credits)

RC1. **Any one of:**
COL778 Principles of Autonomous Systems 3-0 2
COL770 Advanced Artificial Intellegence 3-0-2

Sensing and Perception (SP: 7 credits)

SP1. **JRL780 Computer Vision 3-0-2**
SP2. **ELL705 Stochastic Filtering and Estimation 3-0-0**

Project (P: 20 credits)

P1.**Minor Project (with hands-on experience) 0-0-4**
P2. **Major Project Part-I 0-0-12**
P3. **Major Project Part-II 0-0-24**

The minor project must be a “hands-on” experience. (This includes development of hardware or software or both. A program-wide demonstration will be required.) Significant hands-on experience in Major projects is highly encouraged.

Program Electives (6 credits)

General Electives

JRL880 Special Topics in Robotics I -- 3 0 0
JRL882 Special Topics in Robotics II -- 2 0 0
JRV880 Special Module in Robotics I -- 1 0 0
JRS799 Independent Study – 0 3 0
ELL893 Cyber-Physical Systems (3-0-0)
ELL794 Human-Computer Interface (3-0-0)
ELL800 Numerical Linear Algebra and Optimization in Engineering 3 0 0
ELL706 Optimization for Electrical Engineers 3 0 0
MTL704 Numerical Optimization 3 0 0
MTL729 Computational Algebra and its Applications 3 0 0
ELL788 Computational Perception and Cognition 3 0 0
COL783 Digital Image Analysis 3 0 3
ELL715 Digital Image Processing 3 0 2
COL772 Natural Language Processing 3 0 2
MTL785 Natural Language Processing 3 0 0
ELL720 Advanced Digital Signal Processing 3 0 0
DSL711 Sensors & Transducers 3 0 0
MCL731 Analytical Dynamics 3 0 0
MCL738 Dynamics of Multibody Systems 2 0 2
MCL749 Mechatronics Product Design 3 0 2
MCL837 Advanced Mechanisms 2 0 2
MCL845 Advanced Robotics 2 0 2
ELL700 Linear Systems Theory 3 0 0
MCL741 Control Engineering 3 0 2
ELL702 Nonlinear Systems 3 0 0
ELL703 Optimal Control Theory 3 0 0
ELL704 Advanced Robotics 3 0 0
ELL729 Stochastic Control and Reinforcement Learning 3 0 0
ELL767 Mechatronics 3 0 0
ELL801 Nonlinear Control 3 0 0
ELL802 Adaptive and Learning Control 3 0 0
ELL803 Model Reduction in Control 3 0 0
ELL804 Robust Control 3 0 0
ELL805 Networked and Multi-Agent Control Systems 3 0 0
ELL806 Modeling and Control of Distributed Parameter Systems 3 0 0
ELL807 Stochastic Control 3 0 0
ELL808 Advanced Topics in Systems and Control 3 0 0
MTL811 Mathematical Foundation of Artificial Intelligence 3 0 0
COL770 Advanced Artificial Intelligence 3 0 2
ELL792 Computer Graphics (3-0-0)
COL781 Computer Graphics (3-0-3)
COL774 Machine Learning 3 0 2
ELL784 Introduction to Machine Learning 3 0 0
COL864 Special Topics in Artificial Intelligence 3 0 0
COL870 Special Topics in Machine Learning 3 0 0
ELL789 Intelligent Systems 3 0 0
ELL791 Neural Systems and Learning Machines 3 0 2
ELL795 Swarm Intelligence 3 0 0

ELL882 Large-Scale Machine Learning 3 0 0
ELL888 Advanced Machine Learning 3 0 0
ELL729 Stochastic Control and Reinforcement Learning 3 0 0
ELL787 Embedded Systems and Applications 3 0 0
COL788 Advanced Topics in Embedded Computing (3-0-0)
ELL883 Embedded Intelligence 3 0 0
COL702 Advanced Data Structures 3 0 2
COL758 Advanced Algorithms 3 0 2
COL726 Numerical Algorithms 3 0 2
AIL 721: Deep Learning
AIL 722: Reinforcement Learning
COL785 Virtual and Augmented Reality 3 0 2 [Proposed]
MCL709: Programming Robots with ROS 2-0-2 [Proposed]

Specialization Electives (min. 6 credits in the chosen specialization area)

Collaborative Robotics
MTL768 Graph Theory 3 0 0
MTL763 Introduction to Game Theory 3 0 0
ELL805 Networked and Multi-Agent Control Systems 3 0 0
DSL782 Design for Usability
COL785 Virtual and Augmented Reality

Industrial Robotics
MCL783 Automation in Manufacturing
MCL710 Robotic Automation in Manufacturing 2 0 2 [Proposed]

Rehabilitation and Medical Robotics
MCL798 Medical Robotics [Proposed]
BML830 Biosensor Technology (3 0 2)
BML741 Medical Device Design (2 0 4)
ELL794 Human-Computer Interface

Autonomous and Intelligent Vehicles
P1,P2,P3, should be in Autonomous and Intelligent Vehicles
MCL845 Advanced Robotics 3 0 2
ELL704 Advanced Robotics 3 0 0
CART : Connected and Autonomous Vehicles 3 0 0
CART : Vehicle Telematics 3 0 0
JRL732 Aerial Robotics 3 0 0 [Proposed]

Example Semester-wise plan

Semester I	**FC1 104**	**FC2 300**	**MC1 300**	**BR**		
Semester II	**RC1 302**	**SP1 302**	**SP2 300**	**P1**		
Semester III	**PE1**	**PE2**	**P2**	**MC2 200**	**OC**	
Semester IV			**P3**			